

Quantitative Aptitude

- Q1** The sum of the length and breadth of a cuboid is 40 cm. If the height of the cuboid is $\frac{1}{5}$ of the sum of its length and breadth, then what is the lateral surface area of the cuboid?
 (A) 580cm^2 (B) 320cm^2
 (C) 640cm^2 (D) 720cm^2
- Q2** What is the compound interest on Rs 62,500 for 2 years at 8% per annum compounded yearly?
 (A) Rs 10,600 (B) Rs 10,500
 (C) Rs 10,300 (D) Rs 10,400
- Q3** If a train runs with the speed of 72km/h, it reaches its destination late by 15 minutes. However, if its speed is 90 km/h, it is only 5 minutes late. The correct time to cover its journey in minutes is:
 (A) 45 (B) 35
 (C) 40 (D) 32
- Q4** A mobile phone was sold for ₹ 31,500 after giving two successive discounts of 30% and 10% respectively. What was the marked price of the mobile?
 (A) ₹ 55000 (B) ₹ 50,000
 (C) ₹ 35,000 (D) ₹ 52,500
- Q5** A shopkeeper bought 10 dozen eggs at the rate of ₹5 per egg. 12 eggs broke in transit. He sold the remaining eggs at the rate of ₹6 per egg. Find his profit percentage.
 (A) 10% (B) 20%
 (C) 15% (D) 8%
- Q6** A can complete a work in $11\frac{1}{2}$ days. B is 25% more efficient than A and C is 50% efficient than B. Working together A, B and C will complete the same work
 (A) 3 days (B) 5 days
 (C) 8 days (D) 4 days
- Q7** The angle of depression of a boat from the masthead of a ship 180 m high is 60° . Find the distance (in m) of the boat from the ship.
 (A) 360 (B) $60\sqrt{3}$
 (C) $180\sqrt{3}$ (D) 180
- Q8** The ratio of two numbers is 3 : 4. If the sum of these two numbers is 70 , then what is their LCM?
 (A) 120 (B) 96
 (C) 108 (D) 84
- Q9** A and B can do a piece of work in 16 days and 12 days, respectively. Both work for 4 days and then A goes away. Find how long will B take to complete the remaining work.
 (A) 8 days (B) 7 days
 (C) 6 days (D) 5 days
- Q10** If two numbers are each divided by the same divisor, then the remainders are 6 and 7, respectively. If the sum of the two numbers be divided by the same divisor, then the remainder is 5. The divisor is:
 (A) 13 (B) 8
 (C) 6 (D) 4
- Q11** The price of petrol is increased by 24%. A person wants to increase his expenditure by 14% only. By what percentage (correct to one decimal place), should he decrease his consumption in order to maintain the same level of expenditure?
 (A) 8.3 (B) 7.9
 (C) 8.1 (D) 7.8
- Q12** Pipe A can fill 50% of the tank in 6 hours and pipe B can completely fill the same tank in 18 hours. If both the pipes are opened at the same time, in how much time (in minutes) will the empty tank be completely filled?
 (A) 420 (B) 425
 (C) 432 (D) 435



Q13 The length of each side of a rhombus is 25 m and the length of one of its diagonals is 14 m. Find the area of the rhombus.

- (A) 336 m^2 (B) 175 m^2
(C) 350 m^2 (D) 168 m^2

Q14 A train can cross a pole and a 1000 meters long platform in 5 seconds and 45 seconds respectively. Find the speed of the train.

- (A) 90 km/h (B) 25 km/h
(C) 60 km/h (D) 96 km/h

Q15 If $\theta = 45^\circ$, then what will be the value of $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$?

- (A) 0 (B) 1
(C) -1 (D) ∞

Q16 $6^{25} + 6^{26} + 6^{27} + 6^{28}$ is divisible by:

- (A) 254 (B) 259
(C) 256 (D) 255

Q17 A solid metallic sphere of radius 10 cm is melted and cast into spheres of radius 2 cm each. How many such balls can be made?

- (A) 64 (B) 125
(C) 216 (D) 100

Q18 If sum of squares of zeroes of the polynomial $x^2 + 9x + 3k$ is 21, then what is the value of k ?

- (A) -17 (B) 10
(C) 30 (D) 20

Q19 The length, breadth, and height of a room are 10m, 8m and 6m respectively. Find the cost of white washing the walls of the room and the ceiling at the rate of ₹7.50 per m^2 .

- (A) ₹ 2,220 (B) ₹ 1,850
(C) ₹ 2,150 (D) ₹ 2,000

Q20 The ratio of monthly incomes of Raju and Ankita is 11 : 10 and their monthly expenditures are in the ratio 1 : 2. If Raju saves Rs. 500 per month and Ankita saves Rs. 400 per month, find the monthly income of Ankita?

- (A) Rs. 450 (B) Rs. 500
(C) Rs. 550 (D) Rs. 600

Q21 The smallest number which when increased by 3 and divisible by 27, 35, 25 and 21 is :

- (A) 4728 (B) 4722
(C) 4725 (D) 4720

Q22 The value of

$$\sqrt{32 + \sqrt{11 + \sqrt{21 + \sqrt{8 + \sqrt{64}}}}} \text{ is:}$$

- (A) 6 (B) 8
(C) 7 (D) -6

Q23 Anu fixes the selling price of an article at 25 % above its cost of production. If the cost of production goes up by 20 % and she raises the selling price by 10 %, then her percentage profit is (correct to one decimal place):

- (A) 13.8 % (B) 15.2 %
(C) 14.6 % (D) 16.4 %

Q24 Kishan cycled a distance of 120 km at a certain speed. If he cycled 2 km slower every hour, he would have taken 3 more hours to reach his destination. What is the speed in km/h at which Kishan actually cycled?

- (A) 15 (B) 8
(C) 12 (D) 10

Q25 25 men can complete a task in 16 days. Four days after they started working, 5 more men, with equal workmanship joined them. How many days will be needed by all to complete the remaining task?

- (A) 10 days (B) 15 days
(C) 17 days (D) 19 days

Q26 Vessel A contains milk and water in the ratio 4 : 5 . Vessel B contains milk and water in the ratio 2 : 1. If x litres mixture of A is mixed with y litres mixture of B, then the ratio of milk to water in the mixture becomes 8 : 5. Find the ratio $x : y$.

- (A) 5 : 6 (B) 3 : 10
(C) 2 : 5 (D) 3 : 4

Q27 A thief running at $11\frac{1}{2}$ km /h is chased by a policeman at a speed of $13\frac{1}{2}$ km /h. If the thief



is ahead by 900 m, how much time will the policeman take to catch the thief after starting the chase?

- (A) 22 min 30 sec (B) 27 min
(C) 18 min (D) 4 min 30 sec

Q28 An item is bought on a condition that three equal instalments of ₹3,993 are to be paid at a rate of 10% compound interest, compounded annually. The cost of the item is:

- (A) ₹10,000 (B) ₹9,050
(C) ₹9,590 (D) ₹9,930

Q29 A and B working separately can complete a piece of work in 10 and 16 days, respectively. If they work for a day alternately, with A beginning the work, in how many day(s) will the work be completed?

- (A) $10\frac{1}{4}$ (B) $12\frac{1}{4}$
(C) $1\frac{1}{4}$ (D) $\frac{1}{4}$

Q30 A cylindrical rod of iron, whose height is equal to the radius of its base, is melted and cast into spherical balls, each of whose radius is half the radius of the base of the rod. Find the number of spherical balls thus made.

- (A) 4 (B) 6
(C) 5 (D) 3

Q31 What is the value of

$$\frac{7 \times 4 \div 8}{5 \times 25 \div 125} + \frac{5 \times 4 \div 8}{8 \times 4 \div 16} - \frac{7 \times 4 \div 2}{8 \times 7 \div 4} ?$$

- (A) $\frac{1}{4}$ (B) $\frac{13}{4}$
(C) $\frac{15}{4}$ (D) $\frac{9}{2}$

Q32 What is the value of $\sqrt{150 + \sqrt{256 + \sqrt{9}}}$?

- (A) 14 (B) 15
(C) 13 (D) 17

Q33 ABCD is a cyclic quadrilateral such that AB is a diameter of the circle circumscribing it and $\angle ADC = 118^\circ$, what is the measure of $\angle BAC$?

- (A) 32° (B) 28°
(C) 45° (D) 30°

Q34

In triangle ABC, AD is the bisector of $\angle A$. If $AB = 5$ cm, $AC = 7.5$ cm and $BC = 10$ cm, then what is the distance of D from the midpoint of BC (in cm) ?

- (A) 2.2 (B) 1
(C) 1.5 (D) 2

Q35 The smallest 4-digit prime number is:

- (A) 1009 (B) 1007
(C) 1003 (D) 1001

Q36 The angle of elevation of the top of a tower $25\sqrt{3}$ m high from two points on the level ground on its opposite sides are 45° and 60° . What is the distance in m between the two points (correct to one decimal place)?

- (A) 45.3 (B) 58.4
(C) 68.3 (D) 50.6

Q37 If the mode of the data 21, 17, 21, 19, 17, x is only 21, then the value of x is equal to

- (A) 17
(B) 19
(C) 21
(D) Only natural number

Q38 Simplify : $24 \times (56 - 54) + 256$

- (A) 404 (B) 403
(C) 303 (D) 304

Q39 What must be added to $(x^3 - 3x^2 + 4x - 75)$ to obtain a polynomial that is exactly divisible by $(x - 5)$?

- (A) 4 (B) 6
(C) 2 (D) 5

Q40 If the height and slant height of a cone are 21 cm and 29 cm, respectively. Find its volume, (Use $\pi = \frac{22}{7}$)

- (A) 8800 cm^3
(B) 8708 cm^3
(C) 8440 cm^3
(D) 8080 cm^3

Q41 The cost price of 12 pens is equal to the selling price of 8 pens. The profit percentage is:



- (A) 50% (B) 45%
(C) 40% (D) 55%

Q42 On two items of equal prices, T gets 12% profit on one item and incurs 8% loss on other. What is his overall profit or loss percentage?

- (A) Loss, 4% (B) Profit, 2%
(C) Profit, 4% (D) Loss, 2%

Q43 A specific amount of money is loaned out at a simple interest rate, resulting in ₹15,000 in 2 years and ₹20,000 in 4 years. The rate percentage per annum is:

- (A) 25% (B) 20%
(C) 18% (D) 15%

Q44 The ratio between the present ages of P and Q is 7: 5, respectively. The ratio between P's age 5 years ago and Q's age 5 years hence is 1: 1. What is the ratio between P's age 5 years hence and Q's age 5 years ago?

- (A) 2: 3 (B) 2: 1
(C) 1: 2 (D) 3: 2

Q45 By selling a watch for ₹ 1,600, a person gets profit that is equal to the loss of selling the same watch for ₹ 1,400. If the person wants to get a profit of 20%, at what price should he sell the watch?

- (A) ₹ 1,800 (B) ₹ 1,700
(C) ₹ 1,850 (D) ₹ 1,750

Q46 The fourth proportional of 64,80 and 88 is:

- (A) 90 (B) 100
(C) 120 (D) 110

Q47 A car covers 15 km, 20 km, 30 km and 12 km at speeds of 20 km/h, 30 km/h, 40 km/h and 30 km/h, respectively. The average speed of the car for the total journey is :

- (A) 88 km/h (B) 60 km/h
(C) 30 km/h (D) 40 km/h

Q48 Fourth proportion to 12,18,6 is equal to the third proportion to 4, k. What is the value of k ?

- (A) 6.5 (B) 4

- (C) 6 (D) $4\sqrt{3}$

Q49 If a sum amounts to $2\frac{1}{4}$ times the sum after 2 years on a certain rate of compound interest, compounded annually, then the rate of interest per annum is:

- (A) 30% (B) 50%
(C) 25% (D) 40%

Q50 The sum of the series

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$$

- (A) $\frac{8}{7}$ (B) $\frac{6}{5}$
(C) $\frac{2}{3}$ (D) $\frac{3}{2}$



Answer Key

Q1 (C)
Q2 (D)
Q3 (B)
Q4 (B)
Q5 (D)
Q6 (D)
Q7 (B)
Q8 (A)
Q9 (D)
Q10 (B)
Q11 (C)
Q12 (C)
Q13 (A)
Q14 (A)
Q15 (D)
Q16 (B)
Q17 (B)
Q18 (B)
Q19 (A)
Q20 (B)
Q21 (B)
Q22 (A)
Q23 (C)
Q24 (D)
Q25 (A)

Q26 (B)
Q27 (B)
Q28 (D)
Q29 (B)
Q30 (B)
Q31 (C)
Q32 (C)
Q33 (B)
Q34 (B)
Q35 (A)
Q36 (C)
Q37 (C)
Q38 (D)
Q39 (D)
Q40 (A)
Q41 (A)
Q42 (B)
Q43 (B)
Q44 (B)
Q45 (A)
Q46 (D)
Q47 (C)
Q48 (C)
Q49 (B)
Q50 (D)

CUSTOMER
SERVICE

Hints & Solutions

Q1 Text Solution:

Let the length, breadth and height of the cuboid is x , y and z respectively.

The sum of the length and breadth of a cuboid is 40 cm.

$$x + y = 40$$

The height of the cuboid is $\frac{1}{5}$ of the sum of its length and breadth.

$$z = \frac{1}{5} \times (x + y) = \frac{1}{5} \times 40 = 8 \text{ cm}$$

The lateral surface area of the cuboid is

$$= 2 \times (x + y) \times h$$

$$= 2 \times 40 \times 8$$

$$= 640 \text{ cm}^2$$

Q2 Text Solution:

Given,

Principal (P) = Rs. 62,500

Time (n) = 2 years

Rate (r) = 8%

Compounded interest (CI) = ?

By the using formula,

$$CI = P \left(1 + \frac{r}{100} \right)^n - P$$

$$CI = 62,500 \left(1 + \frac{8}{100} \right)^2 - 62,500$$

$$CI = 62,500 \times \left(\frac{27}{25} \right)^2 - 62,500$$

$$CI = 62,500 \times \frac{729}{625} - 62,500$$

$$CI = 72,900 - 62,500$$

$$CI = 10,400$$

Hence, Option D is the correct answer.

Q3 Text Solution:

If a train runs at the speed of 72 km/h, it reaches its destination late 15 minutes. However, if its speed is 90 km/h, it is only 5 minutes late.

Let, total distance = D km

According to the question,

$$\frac{D}{72} - \frac{D}{90} = (15 - 5) \times \frac{1}{60}$$

$$\frac{D}{72} - \frac{D}{90} = \frac{1}{6}$$

$$\frac{D}{12} - \frac{D}{15} = 1$$

$$15D - 12D = 12 \times 15$$

$$3D = 180$$

$$D = 60 \text{ km}$$

Hence, Distance = 60 km

If the train runs at the speed of 72 km/h, it reaches its destination late 15 minutes.

Therefore, the correct time to cover its journey

$$= \left(\frac{60}{72} - \frac{15}{60} \right) \text{ hour} = \left(\frac{5}{6} - \frac{1}{4} \right) \text{ hour}$$

$$= \left(\frac{7}{12} \right) \text{ hour} = 35 \text{ minutes}$$

Hence, Option B is the correct answer.

Q4 Text Solution:

Given,

$$\text{S.P} = 31500$$

$$\text{Let, M.P} = 100x$$

then,

On giving discount of 30% and 10% successively

$$\text{S.P} = 100x \times \frac{70}{100} \times \frac{90}{100} = 63x$$

Now,

$$63x = 31500$$

$$x = 500$$

Hence, MP of mobile phone =

$$100 \times 500 = \text{Rs. } 50000$$

Hence, the correct answer is (B).

Q5 Text Solution:

Total eggs = 10 dozen = $10 \times 12 = 120$

Cost Price (CP) = $120 \times ₹5 = ₹600$

Eggs sold = $120 - 12 = 108$

Selling Price (SP) = $108 \times ₹6 = ₹648$

Profit = SP - CP = $₹648 - ₹600 = ₹48$

$$\Rightarrow \text{Profit Percentage} = \left(\frac{48}{600} \right) \times 100 = 8\%$$

Hence, Option D is the correct answer.

Q6 Text Solution:

Let, the efficiency of A = $4x$

$$\text{Efficiency of B} = (4x) \times \frac{(100+25)}{100} = (4x) \times \frac{125}{100} = 5x$$

$$\text{Efficiency of C} = (5x) \times \frac{50}{100} = 2.5x$$

$$\text{Total work} = (4x) \times \frac{23}{2} = 46x$$

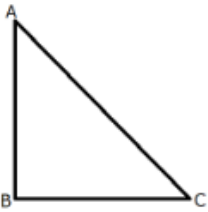
Therefore, required time taken by A, B and C =

$$\frac{46x}{(4x+5x+2.5x)} = \frac{46x}{11.5x} = 4 \text{ days}$$

Hence, Option D is the correct answer.

Q7 Text Solution:





Let, the ship be at point B and the boat is at point C.

AB is the height of the ship.

Angle ACB = 60°

In triangle ABC,

$$\tan 60^\circ = \frac{AB}{BC} \Rightarrow \sqrt{3} = \frac{180}{BC}$$

$$BC = \frac{180}{\sqrt{3}} = 60\sqrt{3} \text{ m}$$

Hence, the correct answer is (B).

Q8 Text Solution:

Let, the two numbers be $3x$ and $4x$

$$3x + 4x = 70$$

$$7x = 70$$

$$x = 10$$

Numbers are 30 and 40

LCM of 30 and 40 is 120

Hence, Option A is the correct answer.

Q9 Text Solution:

Let, the total work (L.C.M of 16,12)= 48 units

$$\text{Efficiency of A} = \frac{48}{16} = 3 \text{ unit}$$

$$\text{Efficiency of B} = \frac{48}{12} = 4 \text{ unit}$$

Total efficiency of A and B= $(3+4) = 7 \text{ unit}$

Work done in 4 days= $(4 \times 7) = 28 \text{ unit}$

Remaining work= $(48-28) = 20 \text{ unit}$

B complete the remaining work= $\frac{20}{4} = 5 \text{ days}$

Hence, Option D is the correct answer.

Q10 Text Solution:

Remainder when the first number is divided by the divisor = 6

Remainder when the second number is divided by the divisor = 7

Remainder when the sum of the two numbers is divided by the same divisor = 5

If the remainders of two numbers when divided by the same divisor are r_1 and r_2 , and the remainder when their sum is divided by the same

divisor is r_3 , then the divisor d can be found using:

$$r_1 + r_2 = r_3 + kd$$

where k is an integer.

Given:

$$r_1 = 6$$

$$r_2 = 7$$

$$r_3 = 5$$

Therefore,

$$6 + 7 = 5 + kd$$

$$\Rightarrow 13 = 5 + kd$$

$$\Rightarrow 13 - 5 = kd$$

$$\Rightarrow 8 = kd$$

Since d must be a divisor of 8, the possible values for d are 1, 2, 4, and 8. However, the only option given that matches this is 8.

Hence, Option B is the correct answer.

Q11 Text Solution:

Let price of petrol be 100 and consumption be x ltr
now total expenditure = $100x$

if price increases by 24% then expenditure will become $124x$

now since we have the increase expenditure only 14% that is $114x$

hence decrease in consumption %

$$\frac{124x - 114x}{124x} \times 100\%$$

$$= \frac{10x}{124x} \times 100\%$$

$$= 8.06\%$$

approx 8.1%

Q12 Text Solution:

Time taken by pipe A fill the 50% of tank = 6hrs

Time taken by pipe A fill the 100% of tank = $6 \times 2 = 12\text{hrs}$

Time taken by pipe B fill the tank = 18 hrs

Now, Portion of tank filled by both pipes together in 1 hour,

$$\Rightarrow \frac{1}{12} + \frac{1}{18} = \frac{3+2}{36}$$

$$\Rightarrow \frac{5}{36}$$

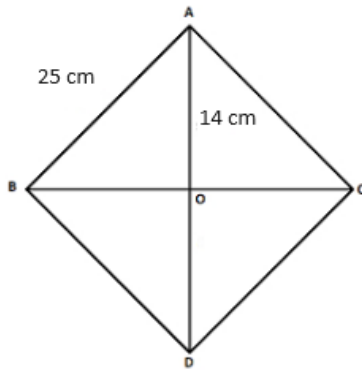
So, time taken to fill the tank when both the pipes are opened together

$$= \frac{36}{5} \text{ hrs} = \frac{36}{5} \times 60 = 36 \times 12 = 432 \text{ minutes}$$



Hence, **Option C** is the correct answer.

Q13 Text Solution:



Let, AD and BC are two diagonals of the rhombus.

$$\therefore AO = \frac{14}{2} = 7 \text{ cm}$$

In AOB right angle triangle,

$$BO = \sqrt{25^2 - 7^2}$$

$$\Rightarrow BO = \sqrt{625 - 49}$$

$$\Rightarrow BO = \sqrt{576}$$

$$\Rightarrow BO = 24$$

$$\therefore BC = (24 \times 2) = 48$$

$\therefore$ The area of the rhombus =

$$\frac{1}{2} \times 14 \times 48$$

$$= 7 \times 48$$

$$= 336 \text{ m}^2$$

Hence, Option A is the correct answer.

Q14 Text Solution:

Given that, a train can cross a pole and a 1000 meters long platform in 5 seconds and 45 seconds respectively.

Let, the speed of the train be 'S' and length be 'L'.

So,

$$S \times 5 = L \dots (i)$$

and

$$S \times 45 = (L + 1000) \dots (ii)$$

From equation (i) and (ii) we get,

$$\Rightarrow \frac{5}{45} = \frac{L}{L+1000}$$

$$\Rightarrow \frac{1}{9} = \frac{L}{L+1000}$$

$$\Rightarrow L + 1000 = 9L$$

$$\Rightarrow 8L = 1000$$

$$\therefore L = 125 \text{ meter}$$

From equation (i)

$$\Rightarrow S \times 5 = 125$$

$$\Rightarrow S = 25 \text{ m/s}$$

$$\Rightarrow S = 25 \times \frac{18}{5} \text{ km/h}$$

$$\therefore S = 90 \text{ km/h}$$

Hence, Option A is the correct answer.

Q15 Text Solution:

$$\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$$

$$\Rightarrow \frac{\sin 45^\circ + \cos 45^\circ}{\sin 45^\circ - \cos 45^\circ}$$

$$\Rightarrow \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}}$$

$$\Rightarrow \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{0}$$

$$\Rightarrow \infty$$

$$\Rightarrow \infty$$

Hence, Option D is the correct answer.

Q16 Text Solution:

Factor out the smallest power of 6:

$$6^{25} (1 + 6 + 6^2 + 6^3)$$

$$\Rightarrow 6^{25} (1 + 6 + 36 + 216)$$

$$\Rightarrow 6^{25} (259)$$

The expression is divisible by 259.

Hence, Option B is the correct answer.

Q17 Text Solution:

Suppose a solid metal sphere of radius 10 cm is melted and cast into n' spheres of radius 2 cm.

Then, volume of metallic sphere of radius 10 cm

= n × volume of sphere of radius 2 cm

$$\Rightarrow \frac{4}{3}\pi(10)^3 = n \times \frac{4}{3}\pi(2)^3$$

$$n = \frac{1000}{8} = 125$$

Hence, 125 spheres of radius 2 cm can be made.

Hence, the correct answer is B.

Q18 Text Solution:

Given,

$$f(x) = x^2 + 9x + 3k$$

The sum of squares of zero = 21

Let, α and β are the zeros of polynomial f(x).

$$\text{Sum of zero } (\alpha + \beta) = -\frac{b}{a} = -\frac{9}{1} = -9$$

$$\text{Product of zero } (\alpha\beta) = \frac{c}{a} = \frac{3k}{1} = 3k$$

Now,



$$(\alpha + \beta) = -9$$

$$\Rightarrow (\alpha + \beta)^2 = 81$$

$$\Rightarrow \alpha^2 + \beta^2 + 2\alpha\beta = 81$$

$$\Rightarrow 21 + 2 \times 3k = 81$$

$$\Rightarrow 6k = 60$$

$$\Rightarrow k = 10$$

Hence, Option B is the correct answer.

Q19 Text Solution:

The room has 4 walls and a ceiling. Let's calculate the surface area of each and then find the total surface area.

There are 2 walls with length 10 m and height 6 m and 2 walls with length 8 m and height 6 m.

Total surface area of walls = $2 \times (\text{length} \times \text{height}) + 2 \times (\text{breadth} \times \text{height})$

$$= 2 \times (10 \times 6) + 2 \times (8 \times 6)$$

$$= 120 + 96 = 216 \text{ m}^2$$

The ceiling has a length of 10m and breadth of 8m.

Surface area of ceiling = length $\times$ breadth

$$= 10\text{m} \times 8\text{m} = 80 \text{ m}^2$$

$$\text{Total surface area to be white washed} = 216 + 80 = 296 \text{ m}^2$$

Cost of white washing = Total surface area $\times$ Rate

$$\text{Cost of white washing} = 296 \times 7.50 = ₹ 2220$$

The cost of white washing the walls of the room and the ceiling at the rate of ₹7.50 per m^2 is ₹2220.

Hence, Option A is the correct answer.

Q20 Text Solution:

Given that, the ratio of monthly incomes of Raju and Ankita is 11 : 10 and their monthly expenditures are in the ratio 1 : 2.

Let, the monthly income of Raju & Ankita be $11x$ & $10x$ respectively.

Let, the monthly expenditure of Raju & Ankita be y & $2y$ respectively.

Since we know that,

Income - Expenditure = Saving

According to the question,

$$11x - y = 500 \dots (i)$$

$$10x - 2y = 400 \dots (ii)$$

Solving equation (i) and (ii) we get,

$$x = 50$$

Therefore, monthly income of Ankita $10x =$

$$10 \times 50 = \text{Rs. } 500$$

Hence, Option B is the correct answer.

Q21 Text Solution:

The prime factorization 27, 35, 25, 21=

$$27 = 3 \times 3 \times 3$$

$$35 = 5 \times 7$$

$$25 = 5 \times 5$$

$$21 = 3 \times 7$$

The LCM of 27, 35, 25, 21=

$$3 \times 3 \times 3 \times 5 \times 5 \times 7 = 4725$$

The smallest number which when increased by 3

and divisible by 27, 35, 25 and 21 is: $(4725 - 3) =$

$$4722$$

Hence, Option B is the correct answer.

Q22 Text Solution:

$$\sqrt{32 + \sqrt{11 + \sqrt{21 + \sqrt{8 + \sqrt{64}}}}}$$

$$= \sqrt{32 + \sqrt{11 + \sqrt{21 + \sqrt{8 + 8}}}}$$

$$= \sqrt{32 + \sqrt{11 + \sqrt{21 + \sqrt{16}}}}$$

$$= \sqrt{32 + \sqrt{11 + \sqrt{21 + 4}}}$$

$$= \sqrt{32 + \sqrt{11 + \sqrt{25}}}$$

$$= \sqrt{32 + \sqrt{11 + 5}}$$

$$= \sqrt{32 + \sqrt{16}}$$

$$= \sqrt{32 + 4}$$

$$= \sqrt{36}$$

$$= 6$$

Hence, the correct answer is (A).

Q23 Text Solution:

Let CP of article = Rs 100

$$\text{SP of article} = 100 \times \frac{125}{100} = \text{Rs } 125$$

$$\text{profit} = 125 - 100 = \text{Rs } 25$$

$$\text{new CP of article} = 100 \times \frac{120}{100} = \text{Rs } 120$$

$$\text{new SP of article} = 125 \times \frac{110}{100} = \text{Rs } 137.5$$

$$\text{New profit} = 137.5 - 120 = \text{Rs } 17.5$$

$$\text{profit \%} = \frac{17.5}{120} \times 100 = 14.6\%$$

Q24 Text Solution:



Let, the actual speed of Kishore's cycle = x km/hr

New speed = $(x - 2)$ km/hr

We know that,

If someone travels a distance at x km/hr speed and then travels the same distance at y km/hr speed, but it takes t hr longer the second time, then the total distance they covered =

$$\frac{xy}{(x-y)} \times t$$

According to the question,

$$120 = \frac{x(x-2)}{x-(x-2)} \times 3$$

$$\Rightarrow 40 = \frac{x^2 - 2x}{x - x + 2}$$

$$\Rightarrow 80 = x^2 - 2x$$

$$\Rightarrow x^2 - 2x - 80 = 0$$

$$\Rightarrow (x - 10)(x + 8) = 0$$

$$x = 10 \text{ or } x = -8$$

$\therefore$ The actual speed = 10 km/hr

Hence, Option D is the correct answer.

Q25 Text Solution:

As we know,

$$\frac{(M1 \times D1)}{W1} = \frac{(M2 \times D2)}{W2}$$

Total work = $25 \times 16 = 400$ units

Work done in four days = $25 \times 4 = 100$ units

Remaining work = $400 - 100 = 300$ units

Now, 5 new men joined.

So, total number of men = $25 + 5 = 30$

Remaining work done by 30 men in x days

$$300 = 30 \times x$$

$$\therefore x = 10$$

Hence, the correct answer is (A).

Q26 Text Solution:

Apply alligation on quantity of milk:

$$\text{Milk in A} = (4/9) \times 117 = 52$$

$$\text{Milk in B} = (2/3) \times 117 = 78$$

$$\text{Milk in mixture} = (8/13) \times 117 = 72$$

Now apply alligation:

$$\text{Difference on B side} = 78 - 72 = 6$$

$$\text{Difference on A side} = 72 - 52 = 20$$

Ratio of quantities of A and B = $6 : 20$

$$\Rightarrow 3 : 10$$

$\therefore$ The ratio is **3 : 10**.

Q27 Text Solution:

Speed of thief = $23/2$ km/h

Speed of policeman = $27/2$ km/h

Distance ahead = 900 m = 0.9 km

Relative Speed = Speed of policeman - Speed of thief

$$\Rightarrow \text{Relative Speed} = (27/2) - (23/2) \text{ km/h}$$

$$\Rightarrow \text{Relative Speed} = 4/2 \text{ km/h} = 2 \text{ km/h}$$

Time (t) = Distance / Relative Speed

$$\Rightarrow t = 0.9 \text{ km} / 2 \text{ km/h}$$

$$\Rightarrow t = 0.45 \text{ hours}$$

$$\Rightarrow t = 0.45 \times 60 \text{ minutes} = 27 \text{ minutes}$$

Hence, Option B is the correct answer.

Q28 Text Solution:

Rate of Interest = 10%

Time = 3 years

Installments = ₹3993

The cost of the item,

$$3993 \left\{ \left(\frac{10}{11} \right) + \left(\frac{10}{11} \right)^2 + \left(\frac{10}{11} \right)^3 \right\}$$

$$= 3993 \times \frac{10}{11} \left\{ \frac{121+110+100}{121} \right\}$$

$$= 3993 \times \frac{10}{11} \times \frac{331}{121}$$

$$= 9930$$

Hence, Option D is the correct answer.

Q29 Text Solution:

Let, the total work be (LCM of 10 and 16) = 80 unit

$$\text{Efficiency of A} = \frac{80}{10} = 8 \text{ unit}$$

$$\text{Efficiency of B} = \frac{80}{16} = 5 \text{ unit}$$

In 2 days they work $(8+5) = 13$ unit

$$\therefore 12 \text{ days work completed} = (13 \times 6) = 78 \text{ unit}$$

$\therefore$ The remaining 2 unit work can be completed in $\frac{2}{8} = \frac{1}{4}$ days

$\therefore$ The total work completed in $12\frac{1}{4}$ days

Hence, Option B is the correct answer.

Q30 Text Solution:

REQUIRED FORMULA:

$$\text{Volume of cylinder is} = \pi r^2 h \text{ unit}^3$$

$$\text{Volume of sphere is} = \frac{4}{3} \pi r^3 \text{ unit}^3$$

GIVEN:

height of cylindrical rod of iron = the radius of its base of cylindrical rod of iron



radius of spherical balls is=half of the radius of the base of the rod.

Let X numbers of spherical balls are casted from cylindrical rod

SOLUTION:

Let height of cylindrical rod is =h unit and radius of its base is r unit

According to the question h=r

Radius of spherical ball is= $\frac{r}{2}$ unit

now,

Volume of spherical balls $\times X$ = volume of cylindrical rod

$$\frac{4}{3}\pi\left(\frac{r}{2}\right)^3 \times X = \pi r^2 h$$

$$\text{or, } X = \frac{24\pi r^2 h}{4\pi r^3} = \frac{24\pi r^3}{4\pi r^3} \quad (\text{since, } r = h) \\ = 6$$

Number of spherical balls thus made are 6

Q31 Text Solution:

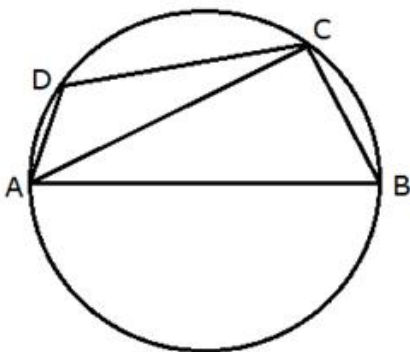
$$\frac{7 \times 4 \div 8}{5 \times 25 \div 125} + \frac{5 \times 4 \div 8}{8 \times 4 \div 16} - \frac{7 \times 4 \div 2}{8 \times 7 \div 4} \\ = \frac{7}{2} + \frac{5}{4} - \frac{14}{14} \\ = \frac{7}{2} + \frac{5}{4} - 1 \\ = \frac{14 + 5 - 4}{4} = \frac{15}{4}$$

Q32 Text Solution:

given-

$$\sqrt{150} + \sqrt{256} + \sqrt{9} \\ = \sqrt{150} + 16 + 3 \\ = \sqrt{169} \\ = 13$$

Q33 Text Solution:



$$\angle ADC = 118^\circ$$

Sum of the opposite angles in cyclic quadrilaterals is 180°

$$\angle ADC + \angle ABC = 180^\circ$$

$$\angle ABC = 180^\circ - \angle ADC = 180^\circ - 118^\circ = 62^\circ$$

Now, the angle subtended by the diameter on the circle is 90°

$$\angle ACB = 90^\circ$$

In $\triangle ABC$,

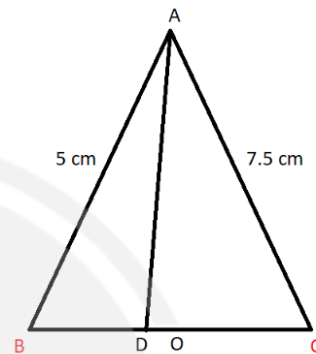
$$\angle BAC = 180^\circ - \angle ABC - \angle ACB = 180^\circ - 62^\circ - 90^\circ = 28^\circ$$

Hence, Option B is the correct answer.

Q34 Text Solution:

In a triangle ABC, AD is the bisector of $\angle A$.

AB= 5 cm, AC = 7.5 cm and BC = 10 cm



Let, AD is the angle bisector of angle A and O is the midpoint of BC.

Using angle bisector theorem :

$$\frac{BD}{DC} = \frac{AB}{AC} = \frac{5}{7.5} = \frac{2}{3}$$

But BC = 10 cm

5 units = 10 cm

1 unit = 2 cm

Hence, BD = 4 cm and DC = 6 cm

As O is the midpoint of BC therefore BO = 5 cm

Hence, DO = BO – BD = 5 cm – 4cm = 1 cm

Hence, Option B is the correct answer.

Q35 Text Solution:

We need to check the primality of the smallest 4-digit numbers.

1000: Not a prime number (divisible by 2, 5).

1001: Not a prime number (divisible by 7, 11, 13).

1002: Not a prime number (divisible by 2).

1003: Not a prime number (divisible by 17, 59).

1004: Not a prime number (divisible by 2).

1005: Not a prime number (divisible by 3, 5).

1006: Not a prime number (divisible by 2).

1007: Not a prime number (divisible by 19, 53).

1008: Not a prime number (divisible by 2, 3).



1009: Check for primality.

1009 is not divisible by any prime number $\leq$

$1009 \div 31.76$

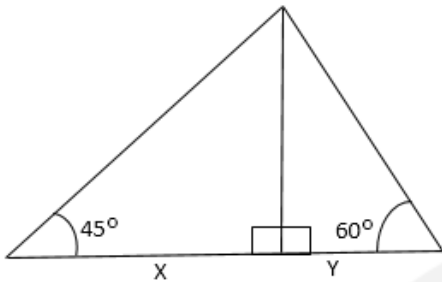
$1009 \approx 31.76$

Divisors checked: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31.

Since 1009 is not divisible by any of these, it is a prime number.

Hence, Option A is the correct answer.

Q36 Text Solution:



Let distance of points from tower be X and Y respectively

So,

$$\tan 45^\circ = \frac{\text{height}}{x} = \frac{25\sqrt{3}}{x}$$

$$1 = \frac{25\sqrt{3}}{x} \quad x = 25\sqrt{3}$$

$$\tan 60^\circ = \frac{\text{height}}{y} = \frac{25\sqrt{3}}{y}$$

$$\sqrt{3} = \frac{25\sqrt{3}}{y}$$

$$y = 25$$

$$X + Y = 25 + 25\sqrt{3}$$

$$= 25 + 43.3 = 68.3$$

Hence, option C is the correct answer.

Q37 Text Solution:

The given numbers are 17, 17, 19, 21, 21, x

Since, the mode of this data is 21. So, 21 should be repeated the maximum number of times.

Now, the frequencies of 17 and 21 are 2 each.

So, the frequency of 21 must be greater than 17.

$\therefore$ The value of x is 21.

Hence, Option C is the correct answer.

Q38 Text Solution:

By using the BODMAS rule; we have

$$24 \times (56 - 54) + 256$$

$$= 24 \times 2 + 256$$

$$= 48 + 256$$

$$= 304$$

Hence, Option D is the correct answer.

Q39 Text Solution:

If a polynomial $f(x)$ is exactly divisible by $(x - a)$, then $f(a) = 0$.

$$\text{Let } f(x) = x^3 - 3x^2 + 4x - 75 + k$$

Since $f(x)$ is divisible by $(x - 5)$, we have $f(5) = 0$.

$$f(5) = 5^3 - 3(5)^2 + 4(5) - 75 + k$$

$$\Rightarrow 125 - 75 + 20 - 75 + k = 0$$

$$\Rightarrow -5 + k = 0$$

$$\Rightarrow k = 5$$

Hence, Option D is the correct answer.

Q40 Text Solution:

We can assume the Height of the cone as the Height of a Right angled Triangle whose base = radius of the cone and slant height represents the Hypotenuse of that Triangle

By Pythagoras theorem,

$$(\text{Slant height})^2 = (\text{Radius})^2 + (\text{Height})^2$$

$$\Rightarrow \text{Radius}^2 = (29)^2 - (21)^2$$

$$\Rightarrow \text{Radius}^2 = (29 + 21)(29 - 21)$$

$$\Rightarrow \text{Radius}^2 = 50 \times 8$$

$$\Rightarrow \text{Radius}^2 = 400$$

$$\Rightarrow \text{Radius} = 20 \text{ cm}$$

Now, the volume of the cone,

$$= \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \times \frac{22}{7} \times 400 \times 21$$

$$= 8800$$

Therefore, the Volume of the cone = 8800 cm^3

Hence, Option A is the correct answer.

Q41 Text Solution:

The cost price of 12 pens is equal to the selling price of 8 pens. The profit percentage is:

Cost price of 12 pen = Selling price 8 pen

Let, the cost price of 1 pen = 10

So, the Cost price of 12 pens = 120

According to the question,

The cost price of 12 pens is equal to the selling price of 8 pens.

So, the selling price of 8 pens = 120.

$$\text{The selling price of each pen} = \frac{120}{8} = 15$$

$$\text{Now selling price of 12 pen} = 15 \times 12 = 180$$

$$\therefore \text{Profit\%} = \frac{(180 - 120)}{120} \times 100 = 50\%$$



Hence, the correct answer is (A).

Q42 Text Solution:

Let the cost price of each item be ₹x.

Profit on the first item = 12% of x = $(12/100) \times x = 0.12x$.

Loss on the second item = 8% of x = $(8/100) \times x = 0.08x$.

Thus, total profit = $0.12x - 0.08x = 0.04x$.

Total cost price = $x + x = 2x$.

Now, overall profit percentage = $(\text{Total profit} \div$

Total cost price) $\times 100$

= $(0.04x \div 2x) \times 100$

= $(0.02) \times 100$

= 2%.

Hence, T's overall profit percentage is 2%.

Hence, Option B is the correct answer.

Q43 Text Solution:

From the given data:

$15000 = P + P \times r \times 2 / 100$

$20000 = P + P \times r \times 4 / 100$

Let us denote the principal amount by P and the rate by r.

$15000 = P (1 + 2r/100)$

$20000 = P (1 + 4r/100)$

Dividing both equations:

$\Rightarrow 20000 / 15000 = (1 + 4r/100) / (1 + 2r/100)$

$\Rightarrow 4 / 3 = (1 + 4r/100) / (1 + 2r/100)$

Cross-multiplying to solve for r:

$\Rightarrow 4 (1 + 2r/100) = 3 (1 + 4r/100)$

$\Rightarrow 4 + 8r/100 = 3 + 12r/100$

$\Rightarrow 4 - 3 = 12r/100 - 8r/100$

$\Rightarrow 1 = 4r/100$

$\Rightarrow r = 100/4$

$\Rightarrow r = 0.25$ or 25%

Hence, Option A is the correct answer.

Q44 Text Solution:

Let, the present age of P and Q be 7x and 5x respectively.

According to the question,

$$\frac{7x-5}{5x+5} = \frac{1}{1}$$

$$\Rightarrow 7x - 5 = 5x + 5$$

$$\Rightarrow 7x - 5x = 5 + 5$$

$$\Rightarrow 2x = 10$$

$$\Rightarrow x = 5$$

$\therefore$ The present age of P and Q = $(7 \times 5) = 35$

and $(5 \times 5) = 25$.

$\therefore$ The required ratio = $(35+5) : (25-5) = 40 : 20 = 2 : 1$

Hence, Option B is the correct answer.

Q45 Text Solution:

Let, the Cost Price of the watch x.

According to the question,

$$1600 - x = x - 1400$$

$$\Rightarrow 2x = 3000$$

$$\Rightarrow x = 1500$$

To get a profit of 20% He has to sell the watch =

$$1500 \times \frac{120}{100} = ₹ 1,800.$$

Hence, Option A is the correct answer.

Q46 Text Solution:

If a, b, c are given, the fourth proportional d is found using the formula:

$$d = (b \times c) / a$$

$$d = (80 \times 88) / 64$$

$$\Rightarrow d = 7040 / 64$$

$$\Rightarrow d = 110$$

Hence, Option D is the correct answer.

Q47 Text Solution:

We know,

$$\text{Average Speed}(s) = \frac{\text{Total Distance covered}}{\text{Total time taken}}$$

$$\therefore s = \frac{15+20+30+12}{\frac{15}{20} + \frac{20}{30} + \frac{30}{40} + \frac{12}{30}}$$

$$\Rightarrow s = \frac{77}{\frac{308}{120}}$$

$$\Rightarrow s = 77 \times \frac{120}{308} = 30 \text{ km/h}$$

Hence, the correct answer is (C).

Q48 Text Solution:

Let, fourth proportion to 12, 18, 6 is x

$$12 : 18 :: 6 : x$$

$$12x = 18 \times 6$$

$$x = 9$$

Fourth proportion to 12, 18, 6 is equal to the third proportional to 4, k.

$$4 : k :: k : 9$$



$$k^2 = 36$$

$$k = 6$$

Hence, Option C is the correct answer.

Q49 Text Solution:

We know that,

$$A = P(1 + r/100)^t$$

Where: A = Amount, P = Principal, r = Rate of interest, t = Time in years

Calculation:

Let the principal = P.

Given, A = (9/4) P and t = 2 years.

Using the compound interest formula:

$$A = P(1 + r/100)^t$$

$$\Rightarrow (9/4)P = P(1 + r/100)^2$$

$$\Rightarrow (9/4) = (1 + r/100)^2$$

$$\Rightarrow \sqrt{(9/4)} = 1 + r/100$$

$$\Rightarrow 3/2 = 1 + r/100$$

$$\Rightarrow r/100 = 3/2 - 1$$

$$\Rightarrow r/100 = 1/2$$

$$\Rightarrow r = 50\%$$

Hence, Option B is the correct answer.

Q50 Text Solution:

$$1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty$$

$$\text{So, sum} = \frac{a}{1-r} = \frac{1}{1-1/3} = \frac{3}{2}$$

Hence the correct answer is option D.



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